STAT 361 Assignment 2 Question 4

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a) Covariance Proof

Proof. By properties of covariance, we know that

$$\begin{aligned} Cov(X,Y) &= E[XY] - E[X]E[Y] \\ &= E[(X - E[X])(Y - E[Y])] \end{aligned}$$

Let $\hat{\beta}_0, \hat{\beta}_1$ be X, Y. From the derivations in the slides, we know:

$$\begin{aligned} \hat{\beta}_1 &= \frac{\sum_{i=1}^n (x_i - \overline{x})(Y_i - \overline{Y})}{\sum_{i=1}^n (x_i - \overline{x})^2} = \sum_{i=1}^n c_i Y_i \\ \hat{\beta}_0 &= \overline{Y} - \hat{\beta}_1 \overline{x} \\ E(\hat{\beta}_1) &= \sum \left(\frac{x_i - \overline{x}}{\sum_{i=1}^n (x_i - \overline{x})^2} \right) (\beta_0 + \beta_1 x_i) = \beta_1 \end{aligned}$$

Let $k = \sum_{i=1}^n (x_i - \overline{x})^2$
Another way to express $Cov(\hat{\beta}_0, \hat{\beta}_1)$ is

$$\begin{aligned} Cov(\hat{\beta}_0, \hat{\beta}_1) &= Cov(\overline{Y} - \hat{\beta}_1 \overline{x}, \hat{\beta}_1) \\ &= Cov(\overline{Y}, \hat{\beta}_1) - Cov(\hat{\beta}_1 \overline{x}, \hat{\beta}_1) \qquad \textbf{(by linearity of covariance)} \end{aligned}$$

Thus, we must derive the two.

Deriving $Cov(\overline{Y}, \hat{\beta}_1)$

Let $\overline{Y} = \sum_{i=1}^n \frac{1}{n} Y_i$ by independence. We know $Var(Y_i) = \sigma^2$, and we simplify

$$\begin{aligned} \hat{\beta}_1 &= \sum_{i=1}^n (x_i - \overline{x})(Y_i - \overline{Y})/k \\ &= \sum_{i=1}^n (x_i - \overline{x}) Y_i / k \end{aligned}$$

since

$$\begin{aligned} \sum_{i=1}^n (x_i - \overline{x})\overline{Y} &= \overline{Y} \underbrace{\sum_{i=1}^n (x_i - \overline{x})}_{0} \\ &= 0 \end{aligned}$$

Let $c_i = (x_i - \overline{x})/k$. By properties of covariance, $Cov(\sum a_i Y_i, \sum b_j Y_j) = \sigma^2 \sum a_i b_i$ where $a_i = \frac{1}{n}$ and $b_i = c_i$ in this context. Thus,

$$\begin{aligned} Cov(\overline{Y}, \hat{\beta}_1) &= \sigma^2 \sum_{i=1}^n \frac{1}{n} c_i \\ &= \sigma^2 \sum_{i=1}^n \frac{1}{n} \left(\frac{x_i - \overline{x}}{k} \right) \\ &= \frac{\sigma^2}{nk} \underbrace{\sum_{i=1}^n (x_i - \overline{x})}_{0} \\ &= 0 \end{aligned}$$

Deriving $Cov(\hat{\beta}_1\overline{x}, \hat{\beta}_1)$

$$\begin{aligned} Cov(\hat{\beta}_1\overline{x}, \hat{\beta}_1) &= \overline{x}\, Cov(\hat{\beta}_1, \hat{\beta}_1) \\ &= \overline{x}\, Var(\hat{\beta}_1) \qquad \textbf{(by property of covariance)} \\ &= \overline{x}\left[\frac{\sigma^2}{k}\right] \end{aligned}$$

Proof Completion

Finally,

$$\begin{aligned} Cov(\hat{\beta}_0, \hat{\beta}_1) &= Cov(\overline{Y}, \hat{\beta}_1) - Cov(\hat{\beta}_1\overline{x}, \hat{\beta}_1) \\ &= 0 - \overline{x}\left[\frac{\sigma^2}{k}\right] \\ &= \frac{-\sigma^2\overline{x}}{\sum_{i=1}^{n}(x_i - \overline{x})^2} \end{aligned}$$

$\therefore$ **We have shown and proved that** $Cov(\hat{\beta}_0, \hat{\beta}_1) = \frac{-\sigma^2\overline{x}}{\sum_{i=1}^{n}(x_i-\overline{x})^2}$

□

(Answer 4a)

b) Normal Distribution Proof

Proof. Let $k = \sum_{i=1}^{n}(x_i - \overline{x})^2$, like part a). Let $s = \sigma^2\left[\frac{1}{n} + \frac{(x-\overline{x})^2}{k}\right] =$. Thus,

$$\hat{Y} = \hat{\beta_0} + \hat{\beta_1}x \sim N(\beta_0 + \beta_1 x, s)$$

We must show the mean is equal to the mean in the normal distribution above, and likewise for the standard deviation.

Mean

From the slides, we know

$$E[\hat{\beta_0}] = \beta_0$$
$$E[\hat{\beta_1}] = \beta_1$$

So it follows that

$$\begin{aligned} E[\hat{Y}] &= E[\hat{\beta_0}] + E[\hat{\beta_1}]x \qquad (x \textbf{ is a constant}) \\ &= \beta_0 + \beta_1 x \end{aligned}$$

Variance

We know $\hat{\beta_0} = \overline{Y} - \hat{\beta_1}\overline{x}$ and we assume $\hat{Y} = \hat{\beta_0} + \hat{\beta_1}x$ holds. Thus,

$$\begin{aligned} \hat{Y} &= \overline{Y} - \hat{\beta_1}\overline{x} + \hat{\beta_1}x \\ &= \overline{Y} + \hat{\beta_1}(x - \overline{x}) \end{aligned}$$

Now we calculate $Var(\hat{Y}) = Var(\overline{Y} + \hat{\beta_1}(x - \overline{x}))$. By properties of variance:

$$Var(\hat{Y}) = Var(\overline{Y}) + Var(\hat{\beta_1}(x - \overline{x})) + 2Cov(\overline{Y}, \hat{\beta_1}(x - \overline{x}))$$

Since $(x - \overline{x})$ is a constant:

$$Var(\hat{Y}) = Var(\overline{Y}) + (x - \overline{x})^2 Var(\hat{\beta_1}) + 2(x - \overline{x})Cov(\overline{Y}, \hat{\beta_1})$$

From a), we know $Cov(\overline{Y}, \hat{\beta_1}) = 0$, so we have:

$$Var(\hat{Y}) = Var(\overline{Y}) + (x - \overline{x})Var(\hat{\beta_1})$$

We know $\overline{Y} = \sum_{i=1}^{n} Y_i/n$ and $Var(\hat{\beta_1}) = \frac{\sigma^2}{k}$ by the slides. And since $Var(Y_i) = \sigma^2$, $Var(\overline{Y}) = Var\left(\sum_{i=1}^{n} \frac{Y_i}{n}\right) = \frac{\sigma^2}{n}$, we have:

$$\begin{aligned} Var(\hat{Y}) &= \frac{\sigma^2}{n} + \frac{\sigma^2}{k}(x - \overline{x}) \\ &= \sigma^2\left[\frac{1}{n} + \frac{(x - \overline{x})}{\sum_{i=1}^{n}(x - \overline{x})^2}\right] \end{aligned}$$

$\therefore$ **We have shown and proved that $\hat{Y}$ has mean $\beta_0 + \beta_1 x$ and variance $= \sigma^2\left[\frac{1}{n} + \frac{(x-\overline{x})}{\sum_{i=1}^{n}(x-\overline{x})^2}\right]$** □

(Answer 4b)